Conserved quantities for the surface quasi-geostrophic equation

Terence Tao · 6 March 2014

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As in the previous post, all computations here are at the formal level only.

In the previous blog post, the Euler equations for inviscid incompressible fluid flow were interpreted in a Lagrangian fashion, and then Noether's theorem invoked to derive the known conservation laws for these equations. In a bit more detail: starting with *Lagrangian space* $\mathcal{L} = (\mathbf{R}^n, \mathrm{vol})$ and *Eulerian space* $\mathcal{E} = (\mathbf{R}^n, \eta, \mathrm{vol})$, we let M be the space of volume-preserving, orientation-preserving maps $\Phi : \mathcal{L} \to \mathcal{E}$ from Lagrangian space to Eulerian space. Given a curve $\Phi : \mathbf{R} \to M$, we can define the *Lagrangian velocity field* $\dot\Phi : \mathbf{R} \times \mathcal{L} \to T\mathcal{E}$ as the time derivative of Φ, and the *Eulerian velocity field* $u := \dot\Phi \circ \Phi^{-1} : \mathbf{R} \times \mathcal{E} \to T\mathcal{E}$. The volume-preserving nature of Φ ensures that u is a divergence-free vector field:

$$\nabla \cdot u = 0. \tag{1}$$

If we formally define the functional

$$J[\Phi] := \frac{1}{2} \int_{\mathbf{R}} \int_{\mathcal{E}} |u(t,x)|^2 \, dxdt = \frac{1}{2} \int_{R} \int_{\mathcal{L}} \left|\dot\Phi(t,x)\right|^2 \, dxdt$$

then one can show that the critical points of this functional (with appropriate boundary conditions) obey the Euler equations

$$[\partial_t + u \cdot \nabla] u = -\nabla p$$

$$\nabla \cdot u = 0$$

for some pressure field $p : \mathbf{R} \times \mathcal{E} \to \mathbf{R}$. As discussed in the previous post, the time translation symmetry of this functional yields conservation of the Hamiltonian

$$\frac{1}{2} \int_{\mathcal{E}} |u(t,x)|^2 \, dx = \frac{1}{2} \int_{\mathcal{L}} \left|\dot\Phi(t,x)\right|^2 \, dx;$$

the rigid motion symmetries of Eulerian space give conservation of the total momentum

$$\int_{\mathcal{E}} u(t,x) \, dx$$

and total angular momentum

$$\int_{\mathcal{E}} x \wedge u(t,x) \, dx;$$

and the diffeomorphism symmetries of Lagrangian space give conservation of circulation

$$\int_{\Phi(\gamma)} u^*$$

for any closed loop γ in $\mathcal{L}$, or equivalently pointwise conservation of the Lagrangian vorticity $\Phi^* \omega = \Phi^* du^*$, where u^* is the 1-form associated with the vector field u using the Euclidean metric η on $\mathcal{E}$, with Φ^* denoting pullback by Φ.

It turns out that one can generalise the above calculations. Given any self-adjoint operator A on divergence-free vector fields $u : \mathcal{E} \to \mathbf{R}$, we can define the functional

$$J_A[\Phi] := \frac{1}{2} \int_{\mathbf{R}} \int_{\mathcal{E}} u(t,x) \cdot Au(t,x)\ dxdt;$$

as we shall see below the fold, critical points of this functional (with appropriate boundary conditions) obey the generalised Euler equations

$$[\partial_t + u \cdot \nabla] Au + (\nabla u) \cdot Au = -\nabla \tilde{p} \tag{2}$$

$$\nabla \cdot u = 0$$

for some pressure field $\tilde{p} : \mathbf{R} \times \mathcal{E} \to \mathbf{R}$, where $(\nabla u) \cdot Au$ in coordinates is $\partial_i u_j A u_j$ with the usual summation conventions. (When $A = 1$, $(\nabla u) \cdot Au = \nabla(\frac{1}{2}|u|^2)$, and this term can be absorbed into the pressure $\tilde{p}$, and we recover the usual Euler equations.) Time translation symmetry then gives conservation of the Hamiltonian

$$\frac{1}{2} \int_{\mathcal{E}} u(t,x) \cdot Au(t,x)\ dx.$$

If the operator A commutes with rigid motions on $\mathcal{E}$, then we have conservation of total momentum

$$\int_{\mathcal{E}} Au(t,x)\ dx$$

and total angular momentum

$$\int_{\mathcal{E}} x \wedge Au(t,x)\ dx,$$

and the diffeomorphism symmetries of Lagrangian space give conservation of circulation

$$\int_{\Phi(\gamma)} (Au)^*$$

or pointwise conservation of the Lagrangian vorticity $\Phi^* \theta := \Phi^* d(Au)^*$. These applications of Noether's theorem proceed exactly as the previous post; we leave the details to the interested reader.

One particular special case of interest arises in two dimensions $n = 2$, when A is the inverse derivative $A = |\nabla|^{-1} = (-\Delta)^{-1/2}$. The vorticity $\theta = d(Au)^*$ is a 2-form, which in the two-dimensional setting may be identified with a scalar. In coordinates, if we write $u = (u_1, u_2)$, then

$$\theta = \partial_{x_1} |\nabla|^{-1} u_2 - \partial_{x_2} |\nabla|^{-1} u_1.$$

Since u is also divergence-free, we may therefore write

$$u = (-\partial_{x_2} \psi, \partial_{x_1} \psi)$$

where the stream function ψ is given by the formula

$$\psi = |\nabla|^{-1} \theta.$$

If we take the curl of the generalised Euler equation (2), we obtain (after some computation) the surface quasi-geostrophic equation

$$[\partial_t + u \cdot \nabla]\theta = 0 \tag{3}$$

$$u = (-\partial_{x_2} |\nabla|^{-1} \theta, \partial_{x_1} |\nabla|^{-1} \theta).$$

This equation has strong analogies with the three-dimensional incompressible Euler equations, and can be viewed as a simplified model for that system; see this paper of Constantin, Majda, and Tabak for details.

Now we can specialise the general conservation laws derived previously to this setting. The conserved Hamiltonian is

$$\frac{1}{2} \int_{\mathbf{R}^2} u \cdot |\nabla|^{-1} u \, dx = \frac{1}{2} \int_{\mathbf{R}^2} \theta \psi \, dx = \frac{1}{2} \int_{\mathbf{R}^2} \theta |\nabla|^{-1} \theta \, dx$$

(a law previously observed for this equation in the abovementioned paper of Constantin, Majda, and Tabak). As A commutes with rigid motions, we also have (formally, at least) conservation of momentum

$$\int_{\mathbf{R}^2} Au \, dx$$

(which up to trivial transformations is also expressible in impulse form as $\int_{\mathbf{R}^2} \theta x \, dx$, after integration by parts), and conservation of angular momentum

$$\int_{\mathbf{R}^2} x \wedge Au \, dx$$

(which up to trivial transformations is $\int_{\mathbf{R}^2} \theta |x|^2 \, dx$). Finally, diffeomorphism invariance gives pointwise conservation of Lagrangian vorticity $\Phi^* \theta$, thus θ is transported by the flow (which is also evident from (3). In particular, all integrals of the form $\int F(\theta) \, dx$ for a fixed function F are conserved by the flow.

— 1. Euler-Lagrange calculations —

We now justify the claim that stationary points of the functional $J_A[\Phi]$ obey (2). We consider continuous deformations $s \mapsto \Phi(s)$ of the critical point Φ, thus Φ now depends on both s and t. We already have the Eulerian velocity field u, which is related to the t derivative of Φ by the formula

$$\partial_t \Phi^i = u^i(\Phi);$$

similarly we may introduce a deformation field v by

$$\partial_s \Phi^i = v^i(\Phi).$$

The vector field v is divergence free and has to obey appropriate vanishing conditions at infinity, but is otherwise unconstrained. If we compute $\partial_t \partial_s \Phi^i = \partial_s \partial_t \Phi^i$ using the above two equations and the chain rule, we arrive at the "zero-curvature" condition

$$\partial_t v^i + u^j \partial_j v^i = \partial_s u^i + v^j \partial_j u^i$$

so in particular

$$\partial_s u^i = \partial_t v^i + u^j \partial_j v^i - v^j \partial_j u^i. \tag{4}$$

On the other hand, as Φ is a critical point, we have

$$\partial_s J_A[\Phi] = 0$$

when $s = 0$. Differentiating under the integral sign and using the self-adjoint nature of A, the left-hand side is

$$\int_{\mathbf{R}} \int_{\mathcal{E}} (\partial_s u^i) A u^i \, dx dt.$$

Inserting (4) and integrating by parts (and using the divergence-free nature of u), this expression can be rewritten as

$$\int_{\mathbf{R}} \int_{\mathcal{E}} v^i (-\partial_t A u^i - u^j \partial_j A u^i - A u^j \partial_i u^j) \, dx dt.$$

Since v^i is essentially an arbitrary divergence-free vector field, the expression inside parentheses must vanish, and the equation (2) follows.

Comments (3)



Anonymous · 6 March 2014

At the end of the second sentence, do you mean 'these equations' instead of 'this equation'.

[Corrected, thanks – T.]

Navier-Stokes Fluid Computers | Combinatorics and more · 25 June 2014

[...] here is a follow up post on Tao's blog (and a few more II, III), and a post on Shtetl [...]



Embedding the SQG equation in a modified Euler equation | What's new · 19 April 2015

[...] where is the active scalar, and is the velocity field. The SQG equations are often used as a toy model for the 3D Euler equations, as they share many of the same features (e.g. vortex stretching); see this paper of Constantin, Majda, and Tabak for more discussion (or this previous blog post). [...]